1 **(a)** $\frac{7}{9}$ **[B1]**

(b) $\frac{36}{99}$ **[M1]**

$\frac{4}{11}$ **[A1]**

[Total Marks: 3]

2 Length multiplier $= \frac{4}{3}$ **[B1]**

Area multiplier $= \frac{7}{6}$ **[B1]**

$\frac{4}{3} \times$ Width multiplier $= \frac{7}{6}$ oe **[M1]**

Width multiplier $= \frac{7}{6} \times \frac{3}{4} = \frac{7}{8}$ **[M1]**

$x = 12.5$ **[A1]**

[Total Marks: 5]

3 **(a)** $100\% - 45\% - 37.5\% = 17.5\%$ oe **[M1]**

$\frac{140}{0.175}$ oe **[M1]**

800 **[A1]**

(b) Junior students $= \frac{3}{8} \times 800 = 300$ **[M1]**

$1.5G + G = 300$ oe **[M1]**

120 **[A1]**

[Total Marks: 6]

4 **(a)** $\frac{75}{1000}$ or equivalent unsimplified fraction **[M1]**

$\frac{3}{40}$ **[A1]**

(b) 3.4 **[B1]**

(c) M1 for converting at least two of the other forms to decimals or percentages correctly (e.g. $\frac{5}{8} = 0.625$, $\frac{7}{11} = 0.636\ldots$, $62\% = 0.62$) **[M1]**

0.6 62% $\frac{5}{8}$ $\frac{7}{11}$ **[A1]**

[Total Marks: 5]

5 **(a)** $\frac{2}{5} = 40\%$ or 0.4, and total percentage of Sales and Production $= 40\% + 28\% = 68\%$ **[M1]**

$100\% - 68\% = 32\%$ and equating $0.32 \times \text{Total} = 24$ or equivalent **[M1]**

75 **[A1]**

(b) Number of women in Sales $= 0.60 \times 30 = 18$ **[M1]**

Number of women in Production $= \frac{3}{7} \times 21 = 9$ (or men $= 12$, so women $= 9$) **[M1]**

Total women $= 18 + 9 + 12 = 39$ **[M1]**

52 **[A1]**

[Total Marks: 7]

6 **(a)** Total number of counters $= x + (2x + 5) + (3x - 1) = 6x + 4$ **[M1]**

$\frac{2x+5}{6x+4} = \frac{35}{100}$ or $\frac{7}{20}$ **[M1]**

$20(2x + 5) = 7(6x + 4)$ leading to $40x + 100 = 42x + 28$ **[M1]**

36 **[A1]**

(b) Substituting $x = 36$ to find green counters $= 3(36) - 1 = 107$ and total counters $= 6(36) + 4 = 220$ **[M1]**

$\frac{107}{220} \times 100$ **[M1]**

$48\frac{7}{11}$ **[A1]**

[Total Marks: 7]